Aurora
FLIGHT SCIENCES
A Boeing Company

IMPLEMENTING FALCON CONTROLLERS AND MISSIONS FOR QUADROTORS

Suchitha Channapatna and Amikosh Dube

INTRODUCTION: SUCHITHA CHANNAPATNA

- Autonomy Division, AI/ML Group
- Entering MS/PhD in Aerospace Engineering
  - MIT, Aerospace Controls Lab (ACL)
- BS in Aerospace Engineering
  - MIT, May 2026
  - Undergraduate research (including at ACL), WORMS (lunar robot)

- Hobbies: singing Carnatic Indian music, learning/teaching Sanskrit language, spending time with friends

Massachusetts Institute of Technology

Aurora FLIGHT SCIENCES
A Boeing Company

INTRODUCTION: AMIKOSH (AVI) DUBE

- Autonomy Division – AI/ML Group
- Masters in ECE (Robotics)
    - Carnegie Mellon University, December 2026
- Bachelors in ECE
    - Purdue University, May 2025
- I have been doing quadcopter research for 6 years
    - Professor. Scherer (CMU)
    - Professor. Sundaram (Purdue)
- Hobbies: Basketball, Film Photos, Storm Chasing, and Astronomy

Carnegie Mellon University

PURDUE UNIVERSITY

AIR LAB

AGENDA

01 FALCON Introduction

02 Suchitha's Work

03 Avi's Work

04 Lessons Learned

05 Q&A

FAST ADAPTATION AND LEARNING FOR CONTROL ONLINE

DARPA LINC

FALCON

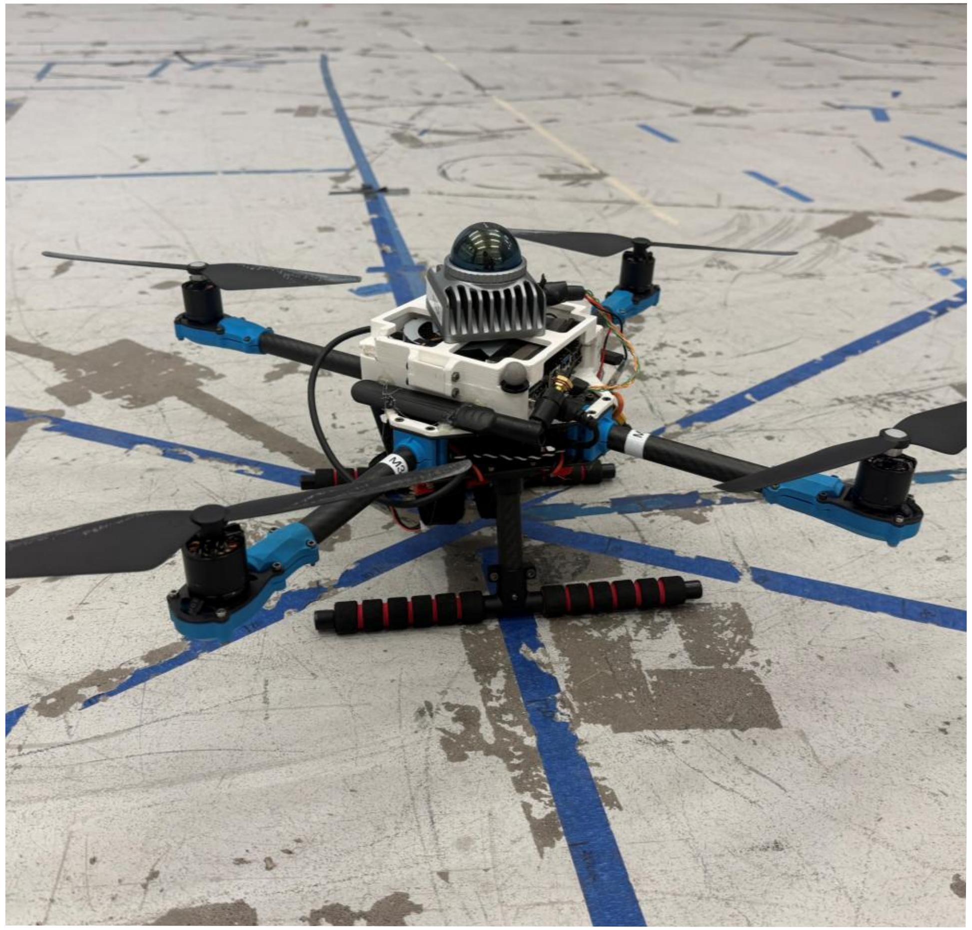
Aurora

MIT

- Adaptive control system to handle unpredicted events and prevent collisions at runtime
- Alleviate burden on operator during unexpected events
    - Currents in the river
    - Motor disturbances
    - FIFA Home Depot bucket

PROJECT GOALS

- Can FALCON be rapidly deployed on several different platforms?
- Intern focus: Quadcopters
  - Collaboration with MIT Aerospace Controls Lab
    - Prof. Jonathan How
    - Kresta Center for Autonomous Systems



SUCHITHA

Relative station keeping mission for quadcopters

SCOPE OF WORK

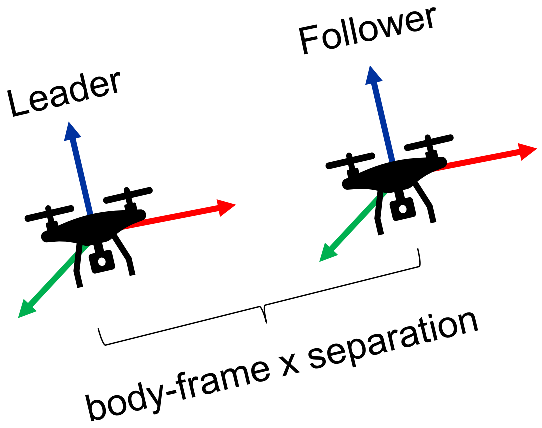
Recap across the internship

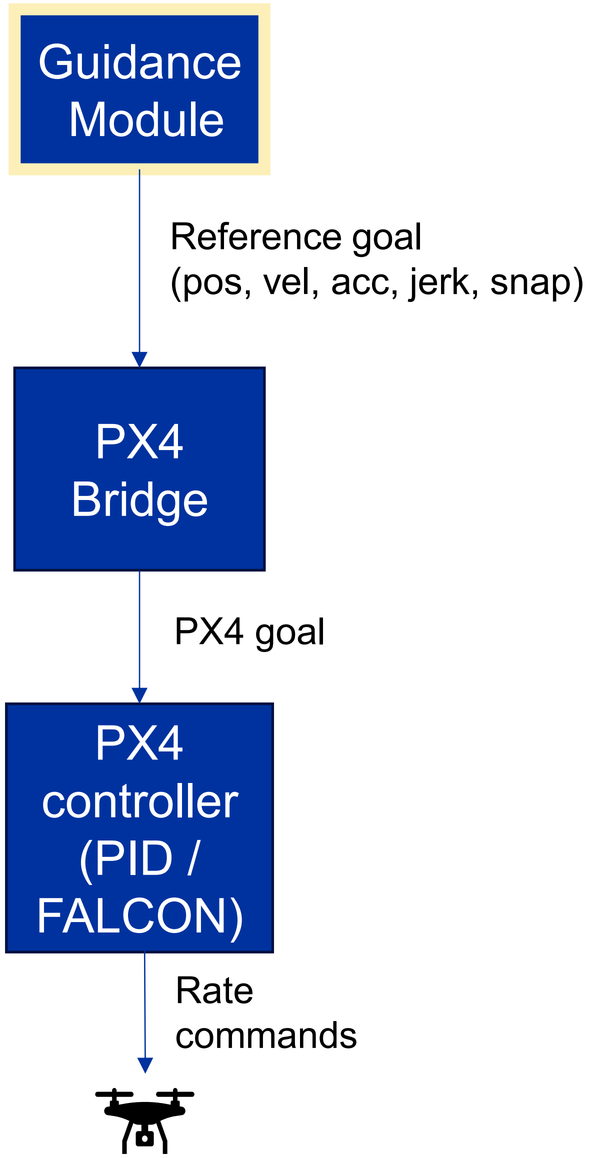
Workstream	Summary	Status
Extension to Multi-vehicle PX4 Sim	Extended ACL’s trajectory generator module and PX4 stack to multi-vehicle simulation	✓ Complete
Visualization Tooling	Built Foxglove tools for Guidance Module and PX4 logs, Python animations/plots for CBF	✓ Complete
Station Keeping Guidance Module	Implemented velocity control law for relative station keeping	✓ Flying on HW
DARPA Demo	Live demonstration of AAOBLTR	✓ Complete
CBF Relative Station Keeping	Closed-form safety filter, validated in simulation and tested on hardware	✓ Flying on HW

RELATIVE STATION KEEPING INTRODUCTION

Goal:
Maintain a **relative separation distance** in **any axes** with respect to the **leader vehicle's body frame**

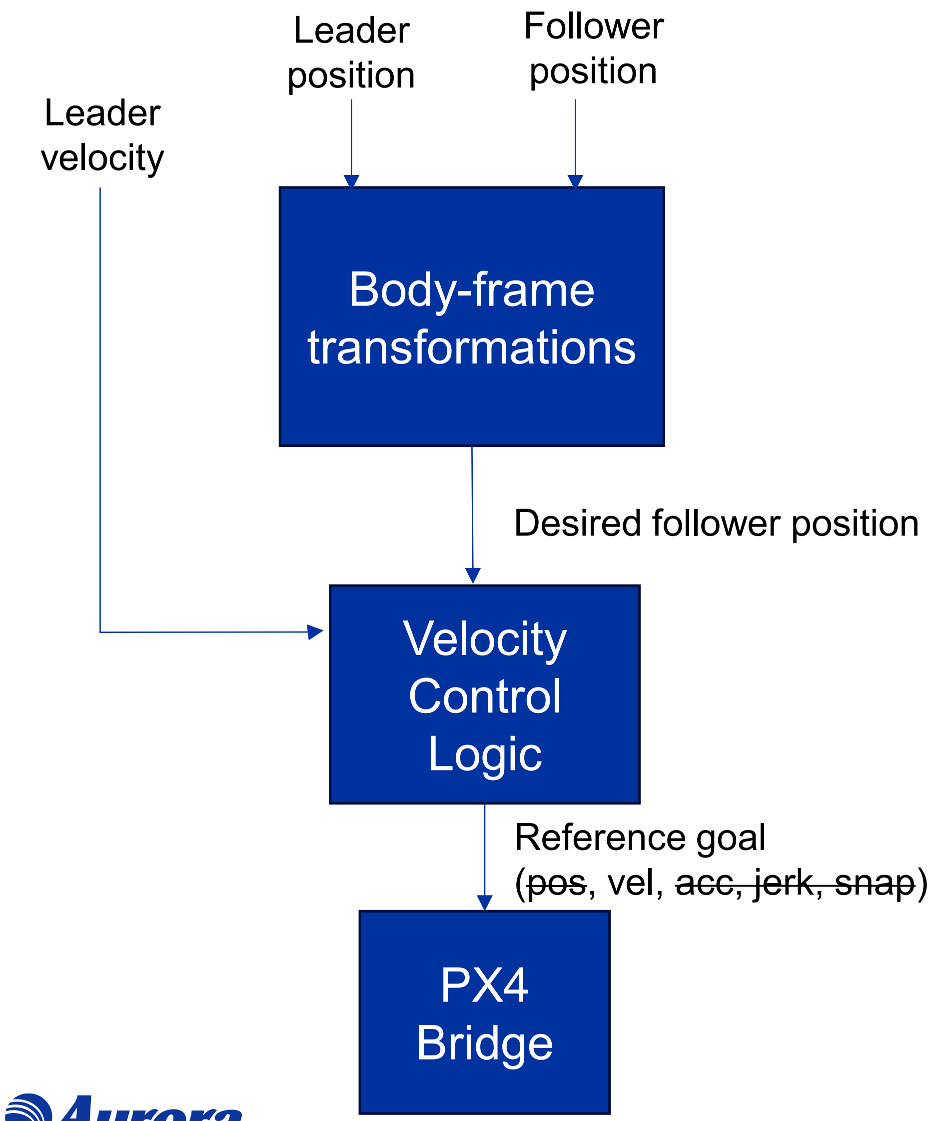
Assumptions:
- Leader's body-x is aligned with yaw
- Follower's yaw always tracks leader's yaw
- Not responsible for deconflicting trajectories

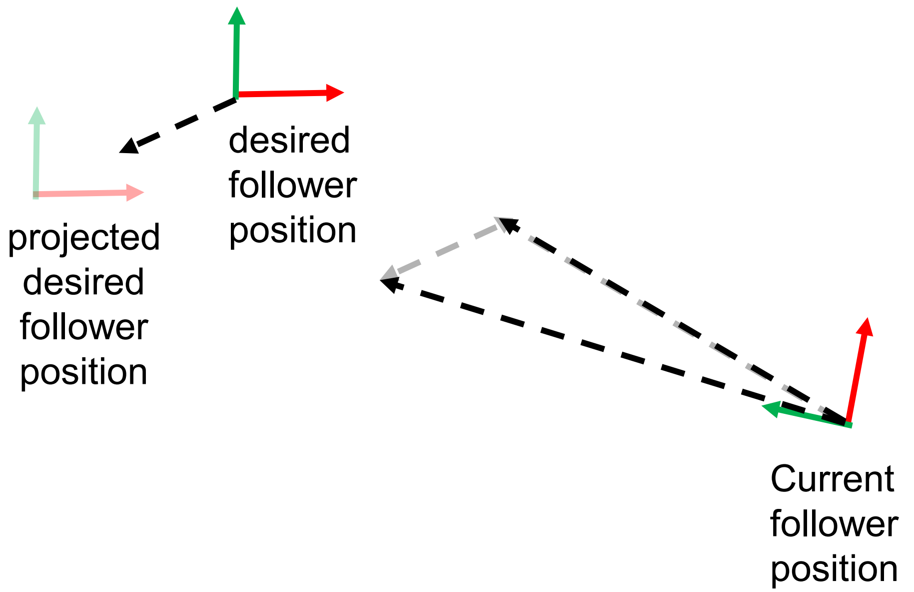




A Boeing Company

RELATIVE STATION KEEPING APPROACH



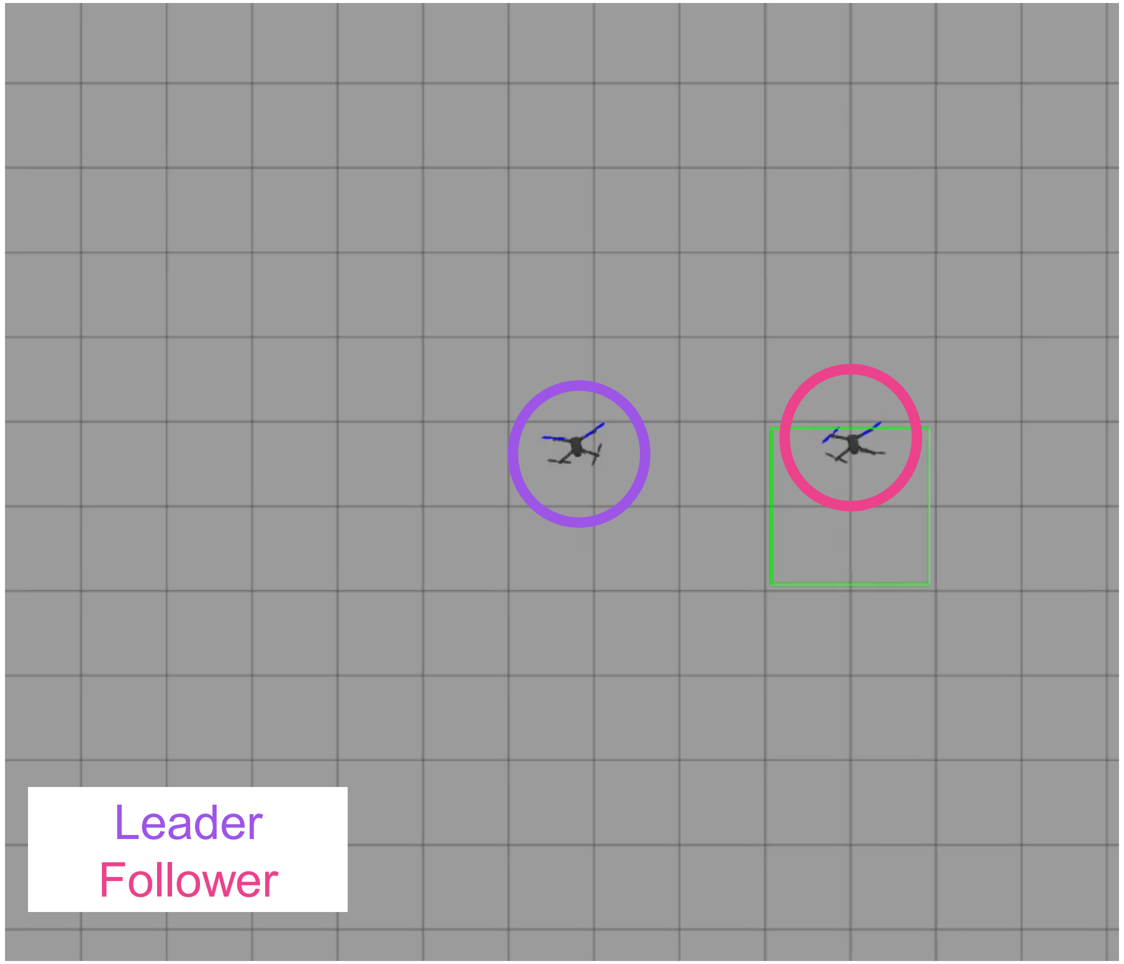


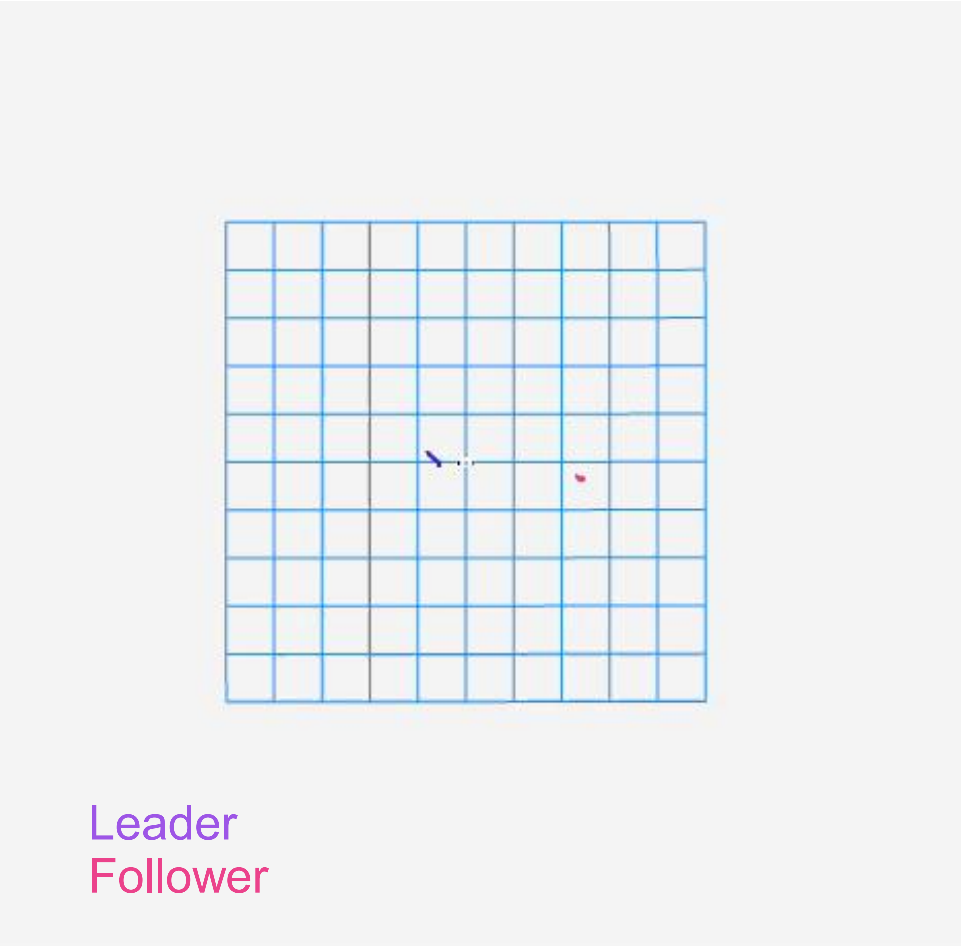
Velocity control law uses leader's velocity and a nonlinear term based on position error.

RELATIVE STATION KEEPING RESULTS - SIMULATION

Leader executes circular trajectory.

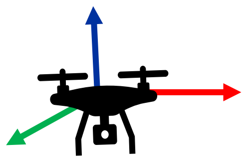
Follower maintains -3.0 meter separation in the leader’s body-frame y-axis **(3 meters right)**.

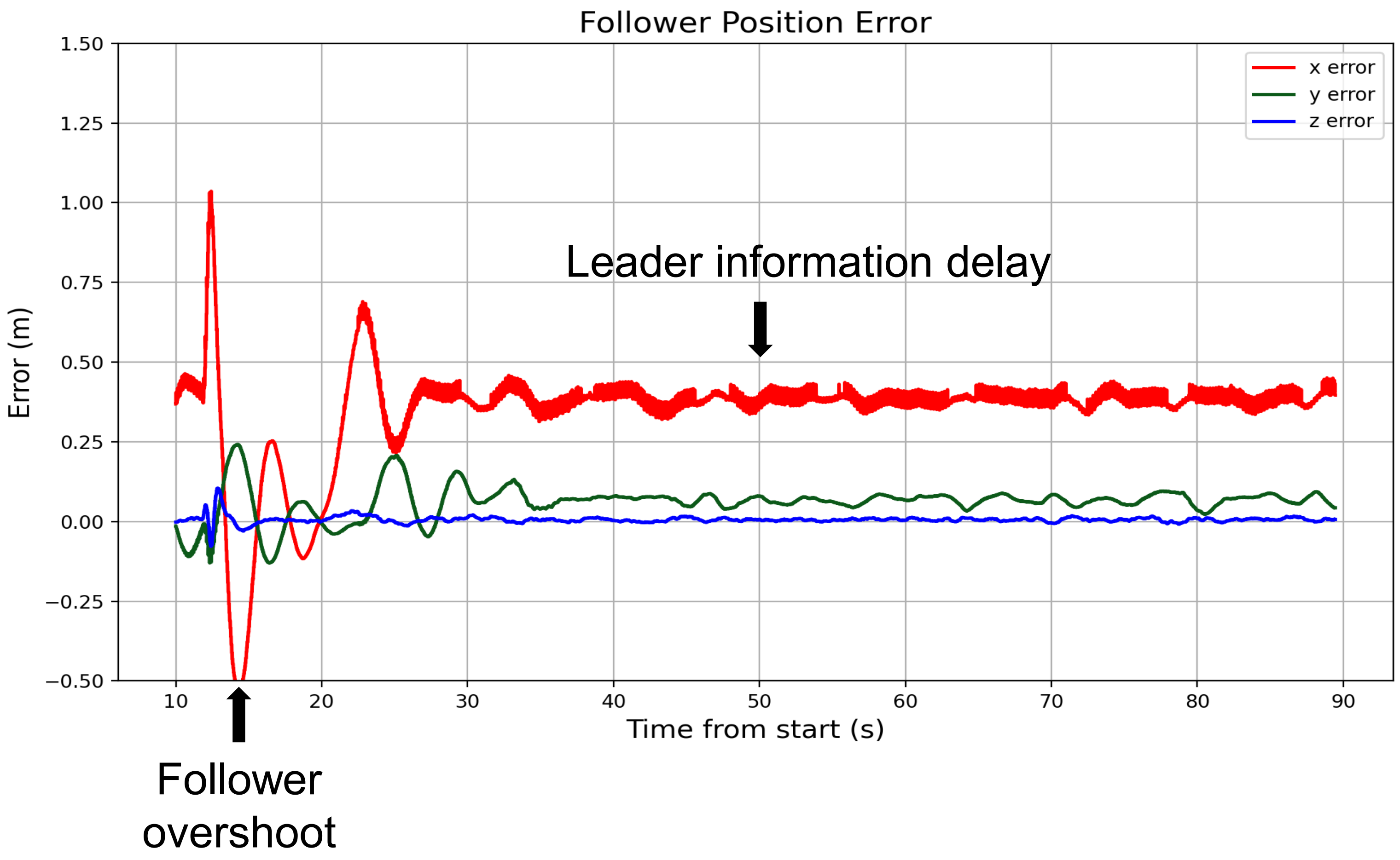




A Boeing Company

RELATIVE STATION KEEPING RESULTS - SIMULATION



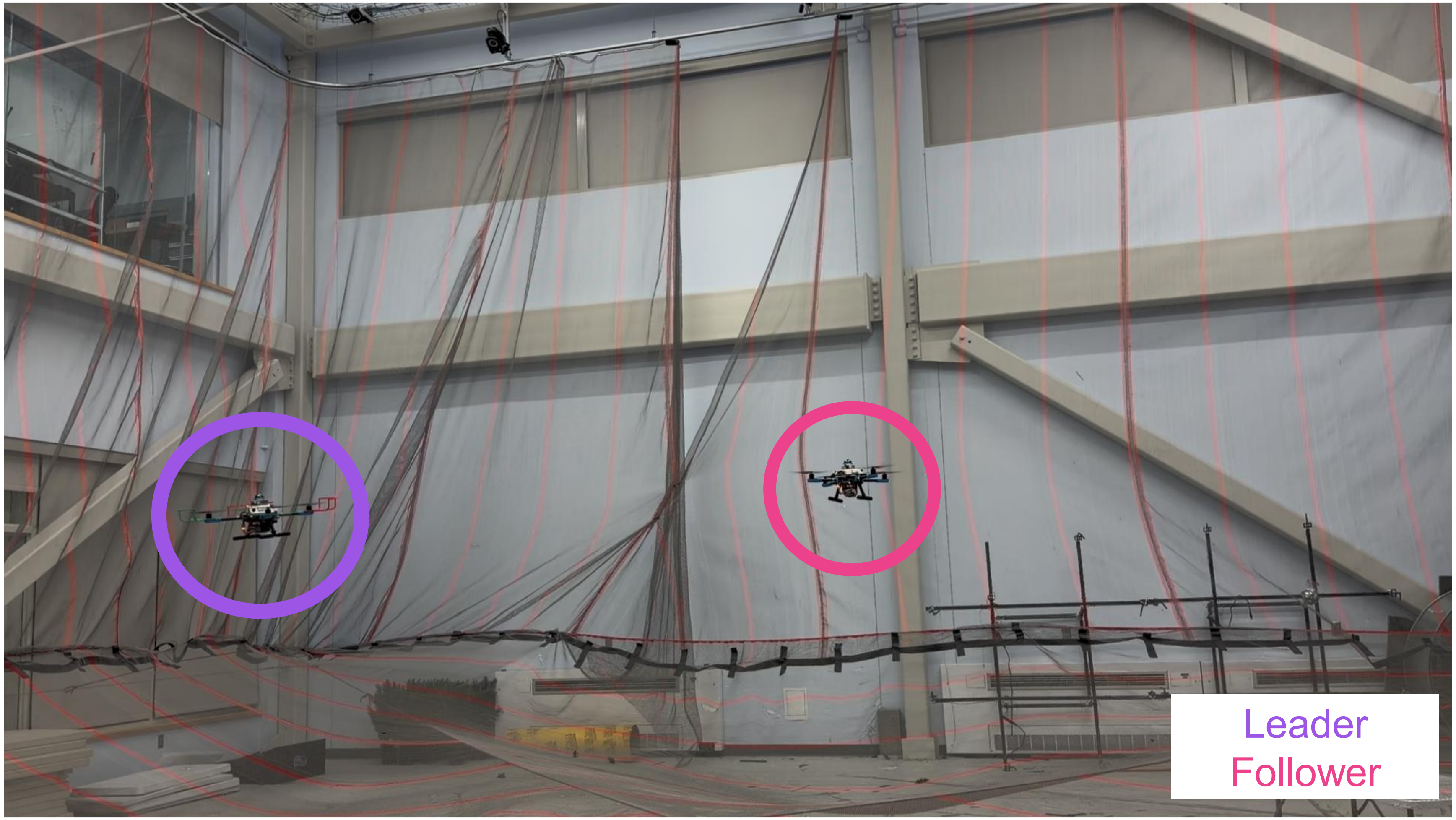


Velocity control law enables tracking. Highest error in x-axis (as expected based on leader trajectory).

RELATIVE STATION KEEPING VIDEO - HARDWARE

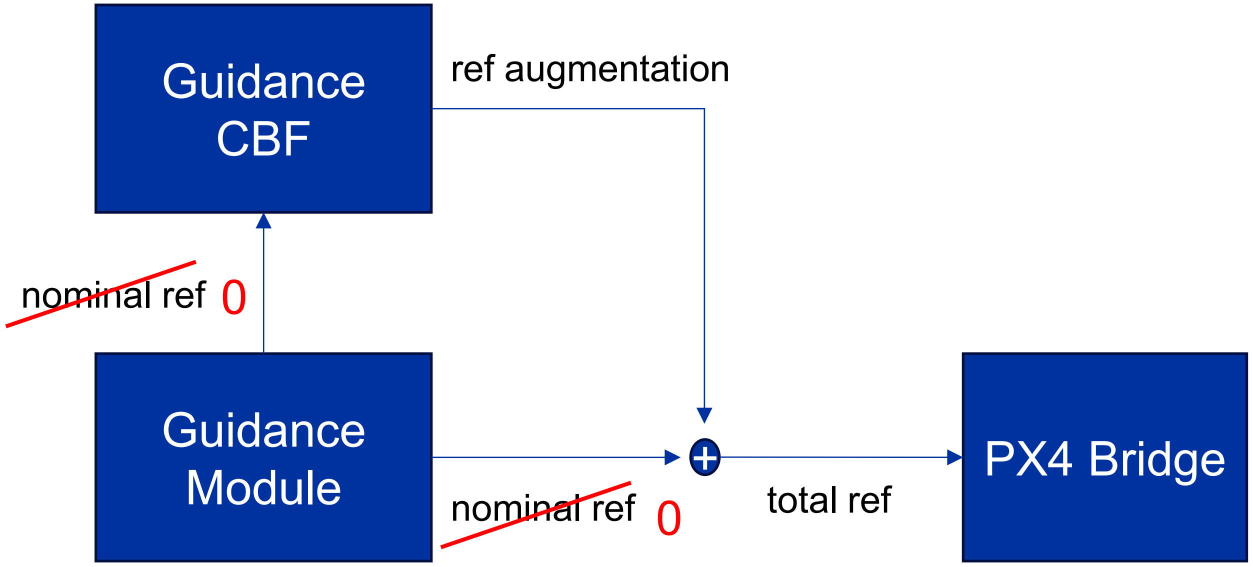
Leader is hand-flown.

Follower maintains -3.0 meter separation in leader's body-frame y-axis **(3 meters right)**.



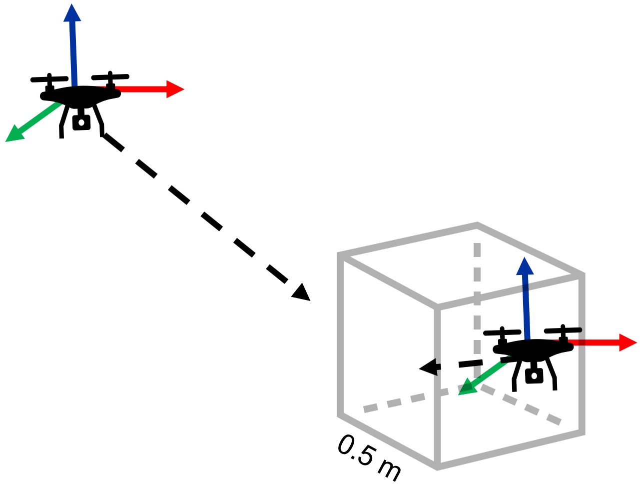
EXPORT CONTROLLED - This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR

CONTROL BARRIER FUNCTION - STATION KEEPING



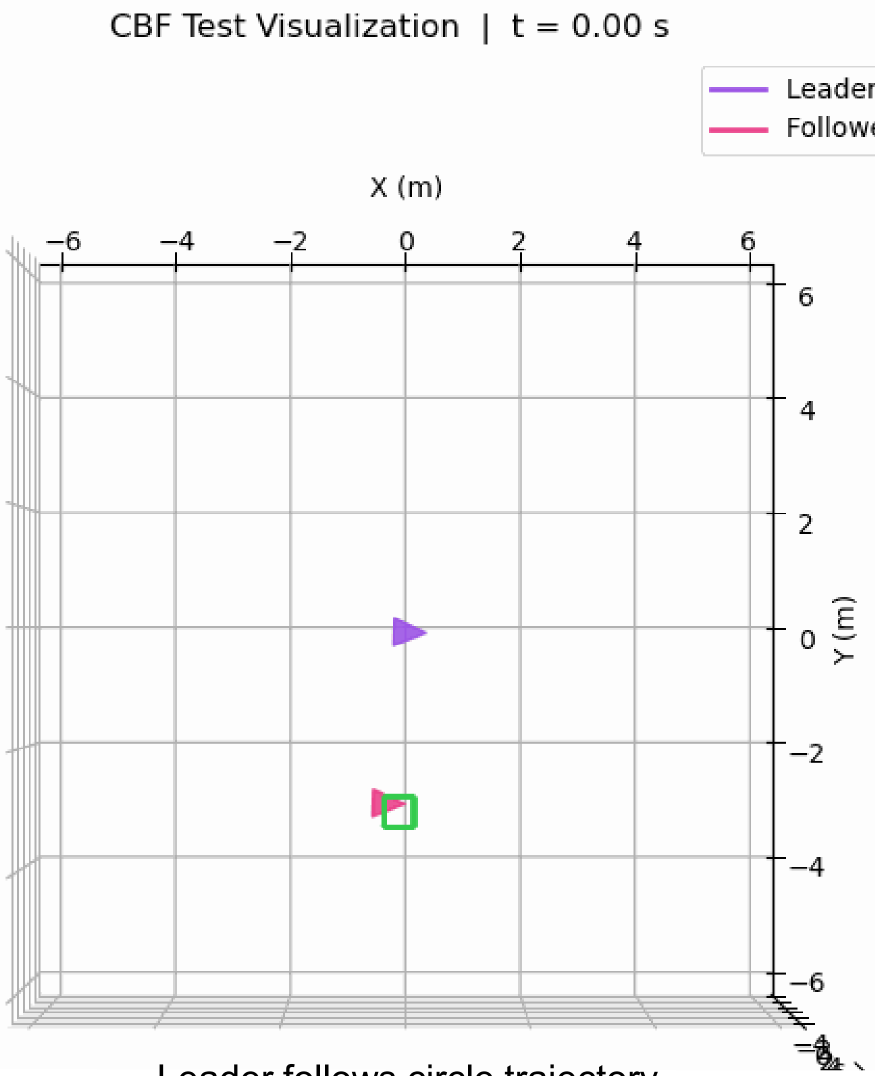
Analytical CBF solution possible for **box constraints**

$\dot{p} = v$



Safe set centered at follower's desired position

CBF FOR STATION KEEPING RESULTS - SIMULATION



Leader follows circle trajectory.
Follower tracks **3 m to the right. No nominal velocity commands** used (CBF only).

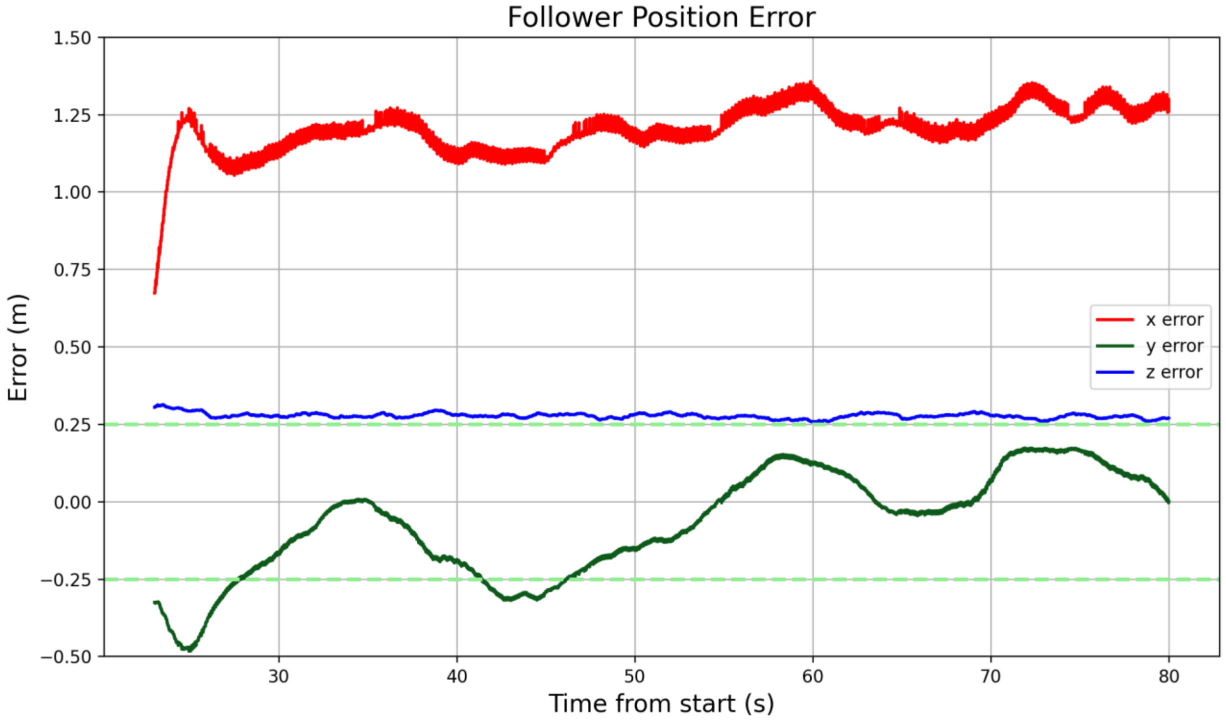
CBF was successfully used for tracking, but follower did not remain within moving barrier during trajectory.

Potential reasons:

- Over-simplified model representation
- Tuning of CBF parameter
- Size of safe set
- Changing heading (better results with Line trajectory)

Aurora FLIGHT SCIENCES
A Boeing Company

CBF FOR STATION KEEPING RESULTS - SIMULATION



CBF without nominal reference command

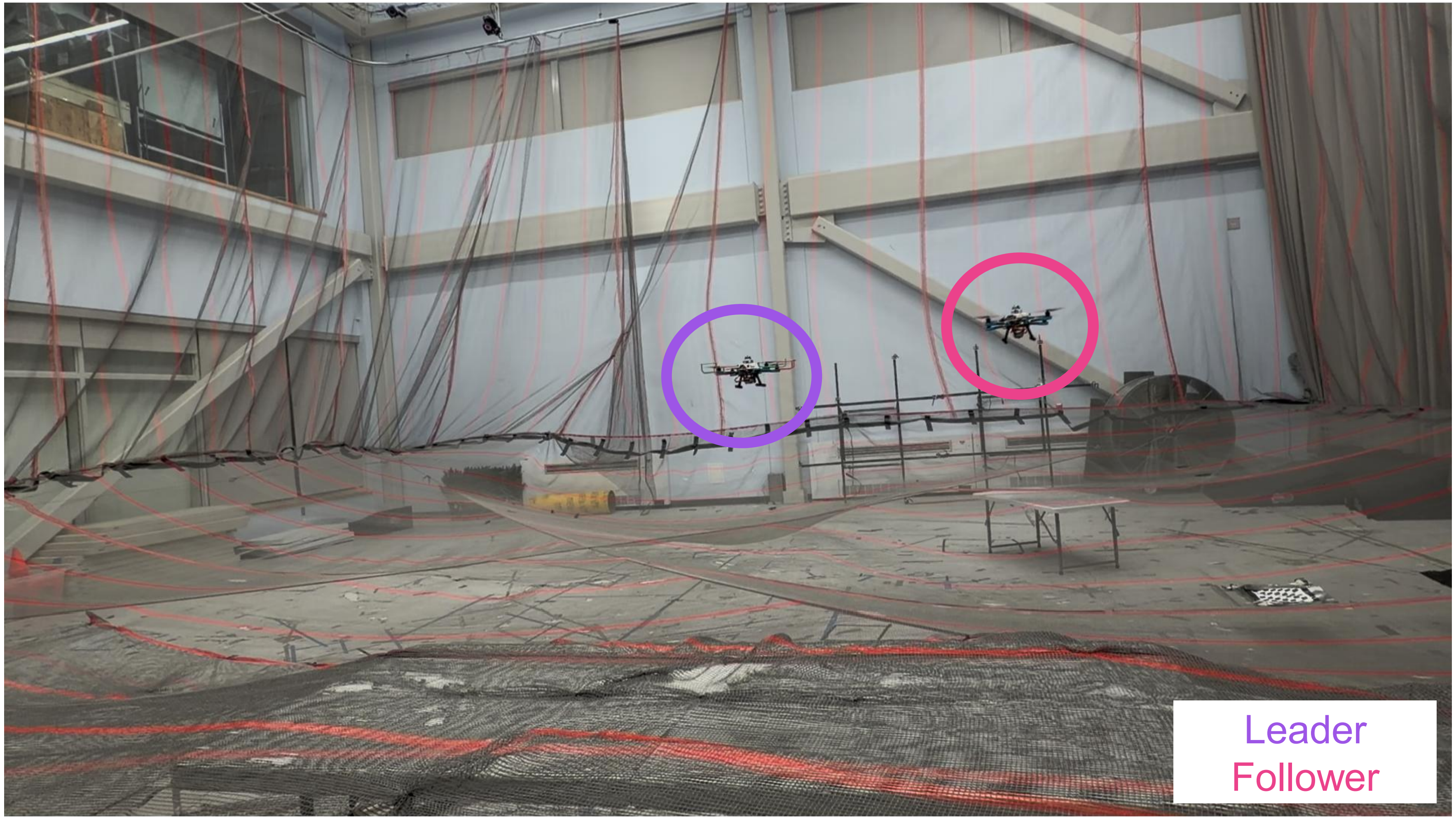
CBF with nominal reference command

x-error violates CBF constraint.
x-error reduces when nominal velocity command is used.

CBF FOR STATION KEEPING VIDEO - HARDWARE

Leader is hand-flown.

Follower maintains -3.0 meter separation in leader's body-y axis **(3 meters right)**.



LESSONS LEARNED

Technical

- *Effectively* going down a PX4 rabbit-hole
- Sim → Real gap
    - Aligning simulation with hardware (and when it is unnecessary)
- Simplifying development and testing
    - Modular architectures and configuration files

Professional

- Preparing for a formal demonstration
    - Communication, automation, clarity of demo goals

AVI

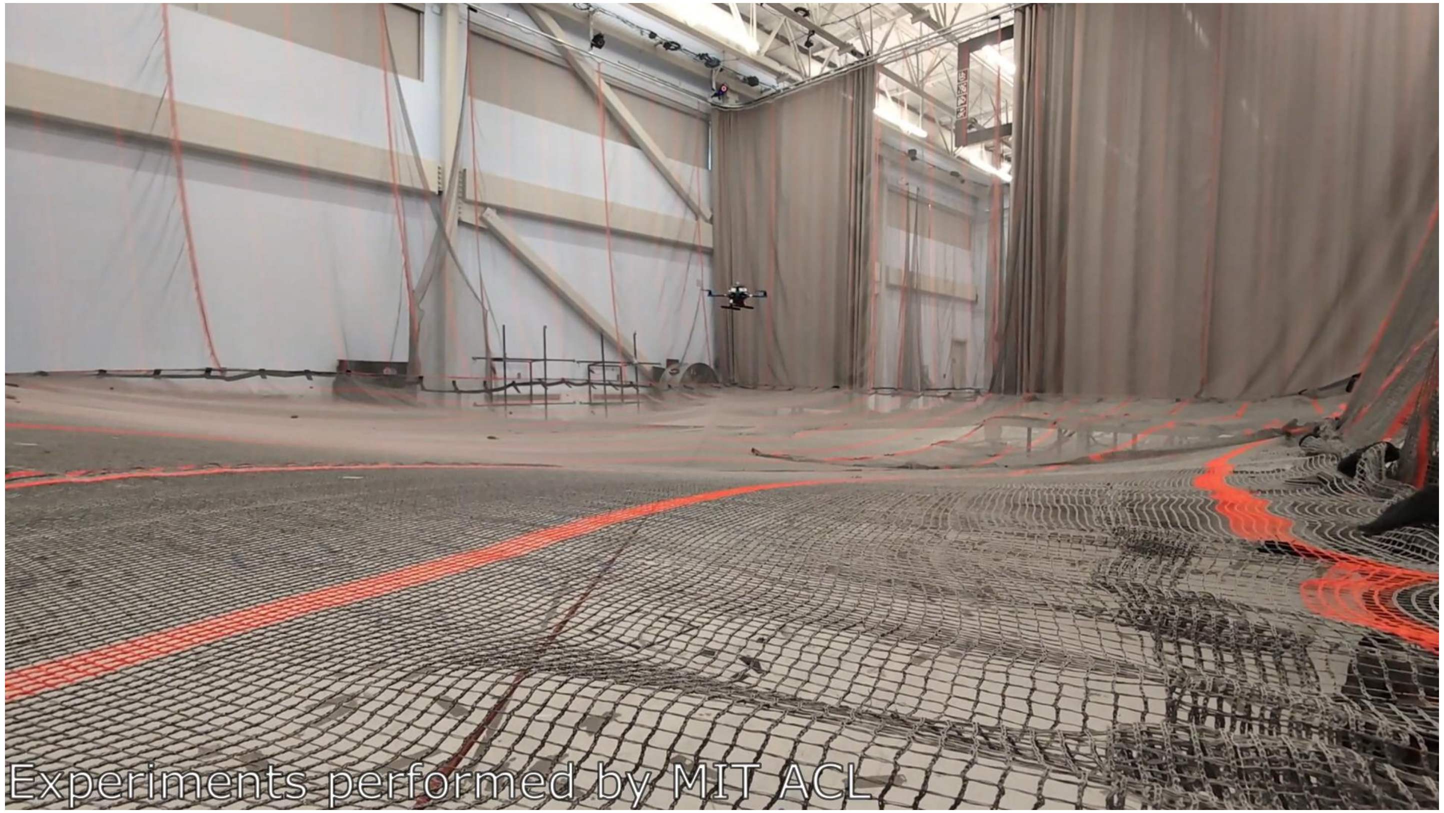
FALCON controllers implemented for quadcopters

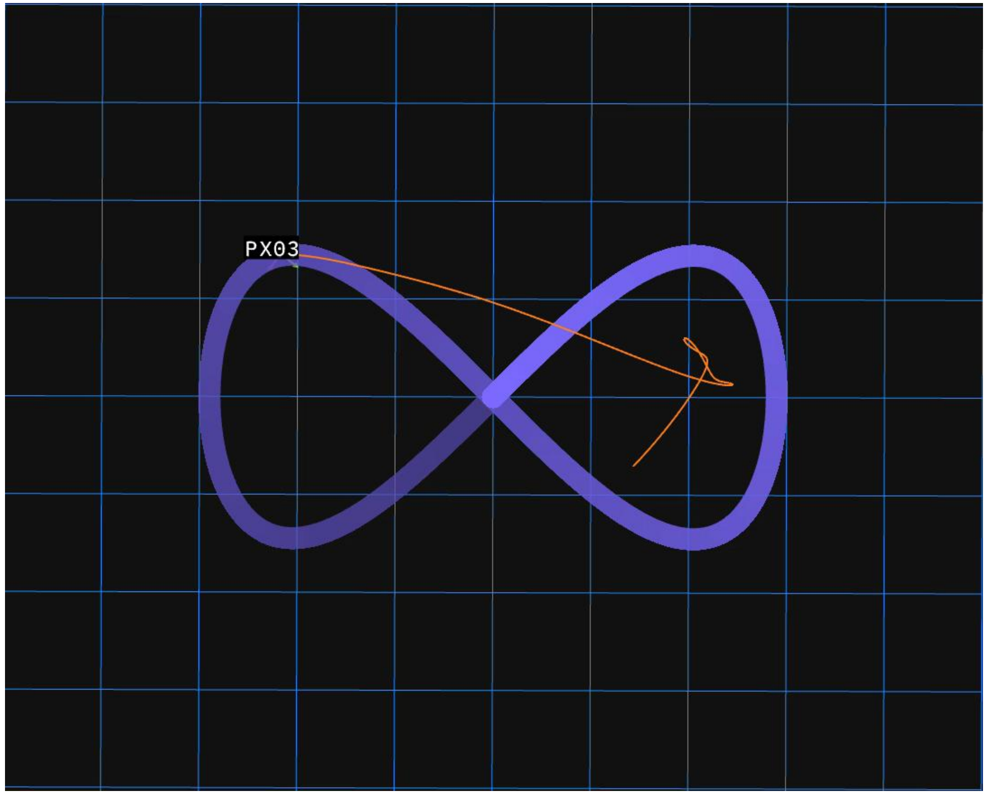
SCOPE OF MY WORK

Recap across the internship

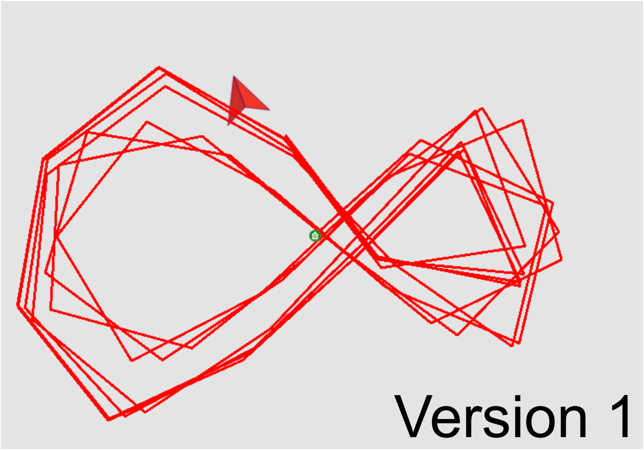
Workstream	Summary	Status
PX4 Debugging	Diagnosed and resolved issue in prior figure-8 implementation	✓ Fixed
Analyzation Tooling	Built Foxglove tools, iterated RViz → full 3D replay	✓ Complete
Controller Library	Implemented PID, RSLQR, OBLTR, AAOBLTR behind a swappable interface	✓ Flying on HW
AAOBLTR Tuning	Tuned adaptive observer-based controller	✓ Stable flight
DARPA Demo	Live demonstration of AAOBLTR	✓ July 22
CBF Obstacle Avoidance	Closed-form safety filter, validated in simulation	⚠ Sim only — HW blocked
HLOC Perception	Improved boat detection / pose estimation pipeline	✓ Accuracy improved

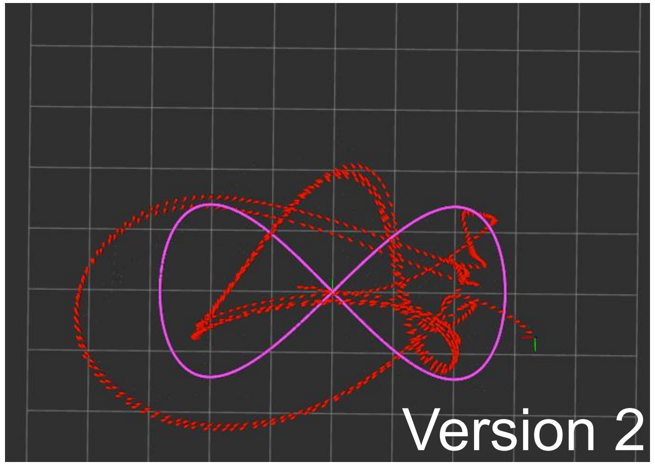
DEBUGGING PREVIOUS FALCON IMPLEMENTATION

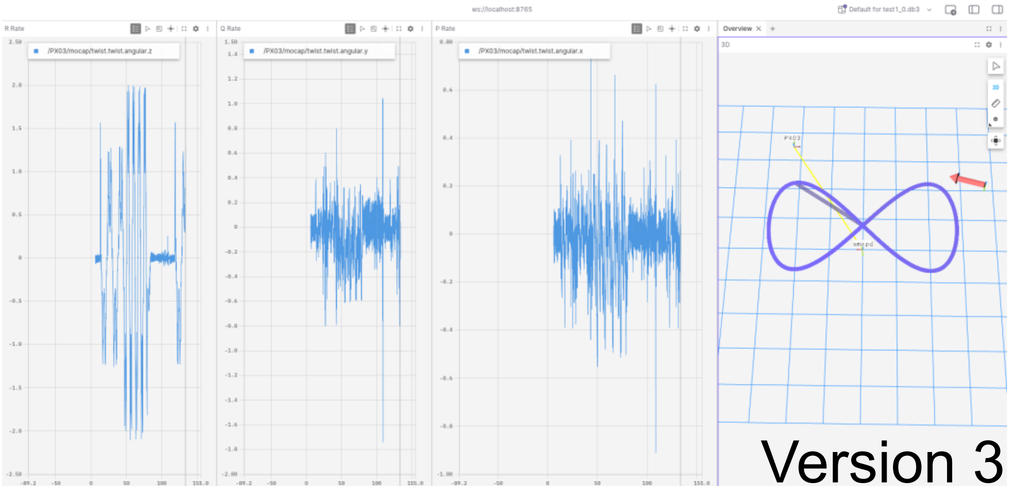


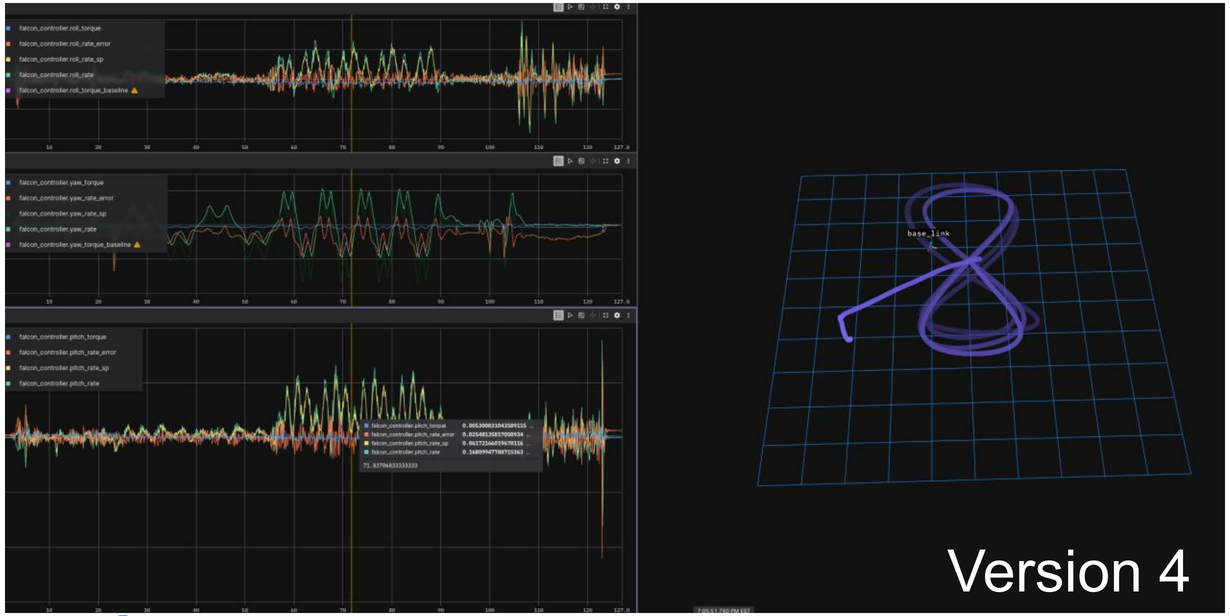


FOXGLOVE ANALYZATION TOOLS



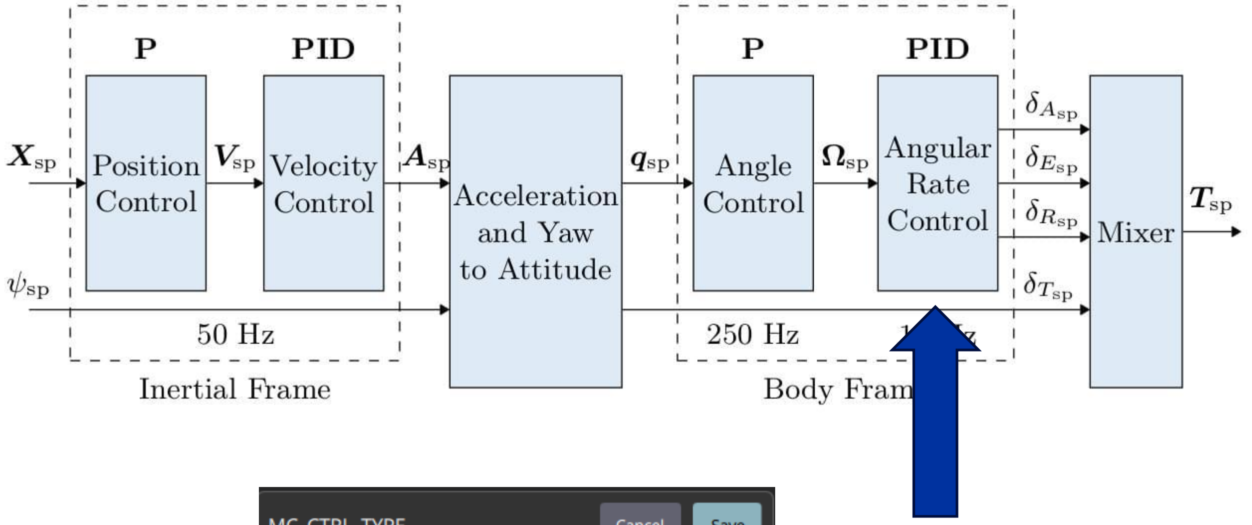


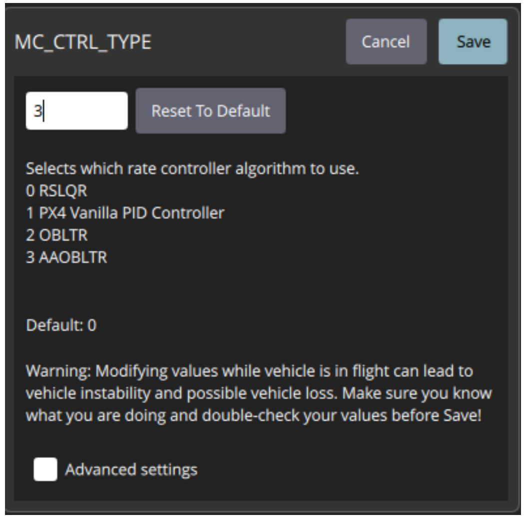




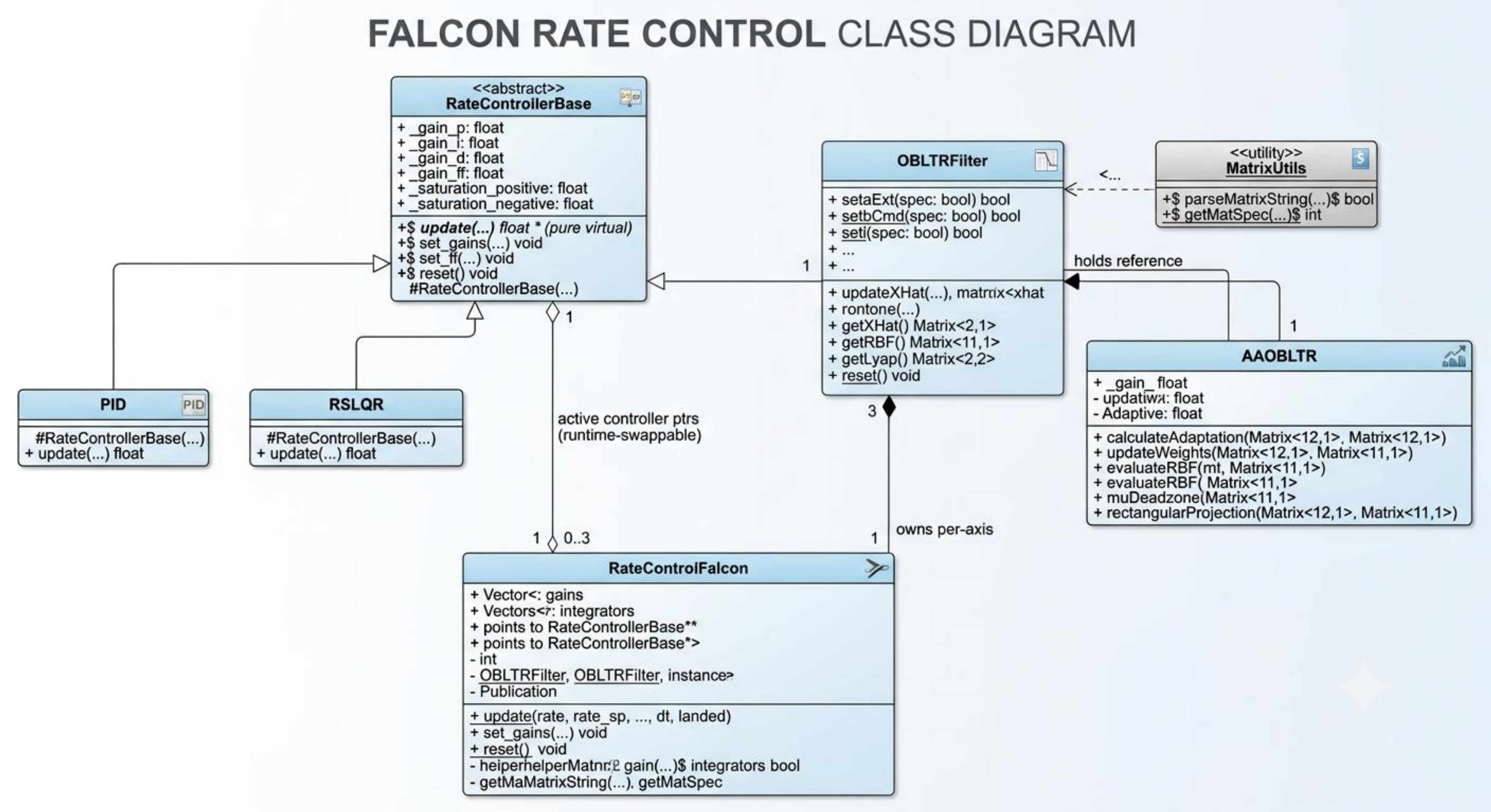
CUSTOM RATE CONTROLLER LIBRARY IN PX4

- Implemented in the angular rate control loop
- Added new uORB message types
- Extended QGC with custom tuning fields
  - Can tune in the field like with PID
  - Can not adjust Q & R





CUSTOM RATE CONTROLLER LIBRARY IN PX4



A Boeing Company

ADAPTIVE AUGMENTED OBSERVER-BASED CONTROL WITH LOOP TRANSFER RECOVERY (AAOBLTR)

- OBLTR:
  - Estimates states where the drone can't directly sense due to noise or additional sensors
  - Flies as if every state were known
- AAOBLTR:
  - Builds an OBLTR with online self-adjustment
  - Converges back to designed transient performance

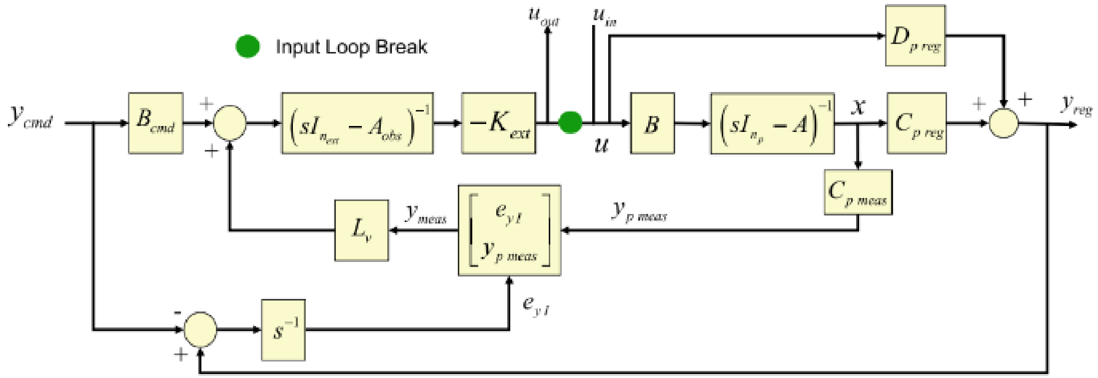
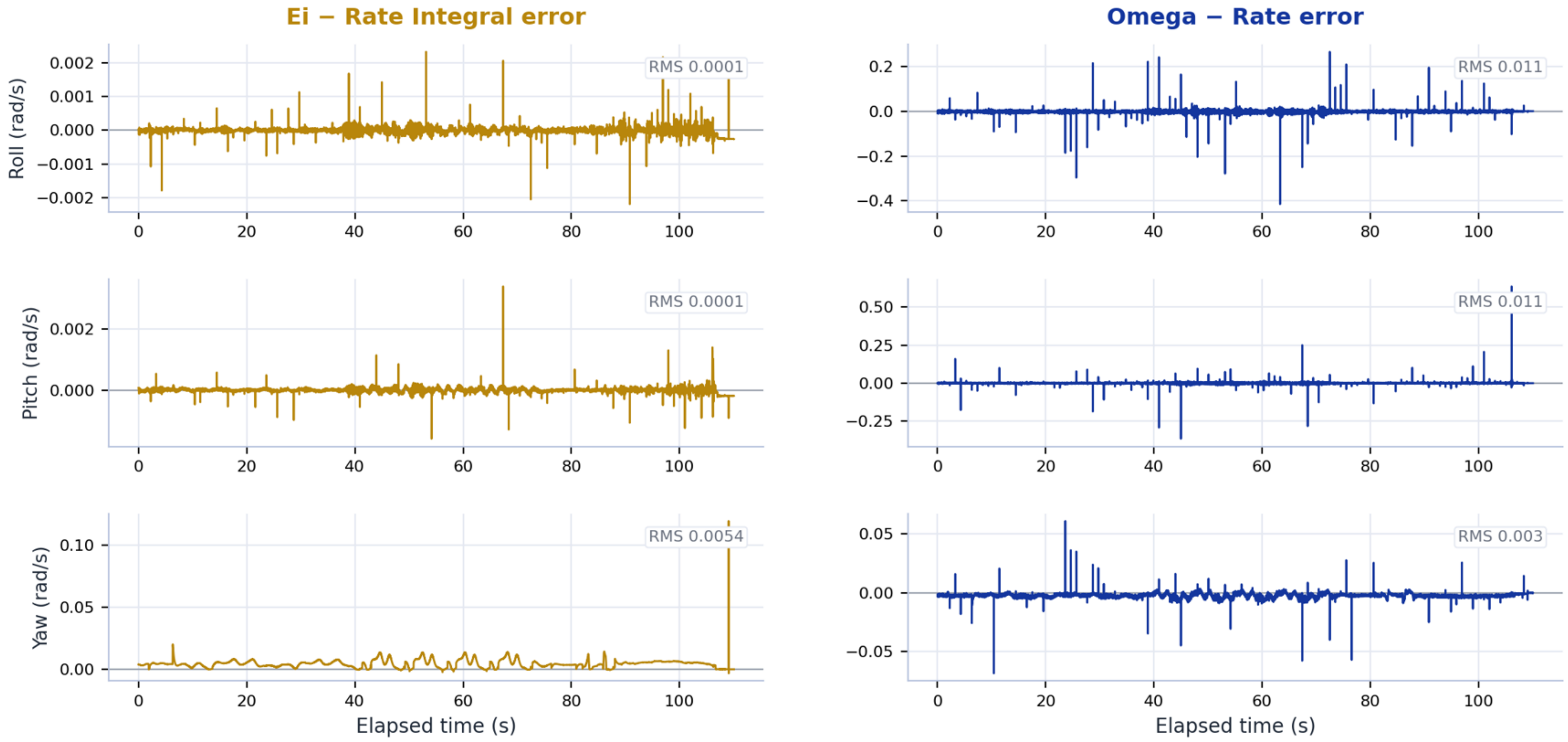


Fig. 6.13 Input loop gain calculation block diagram

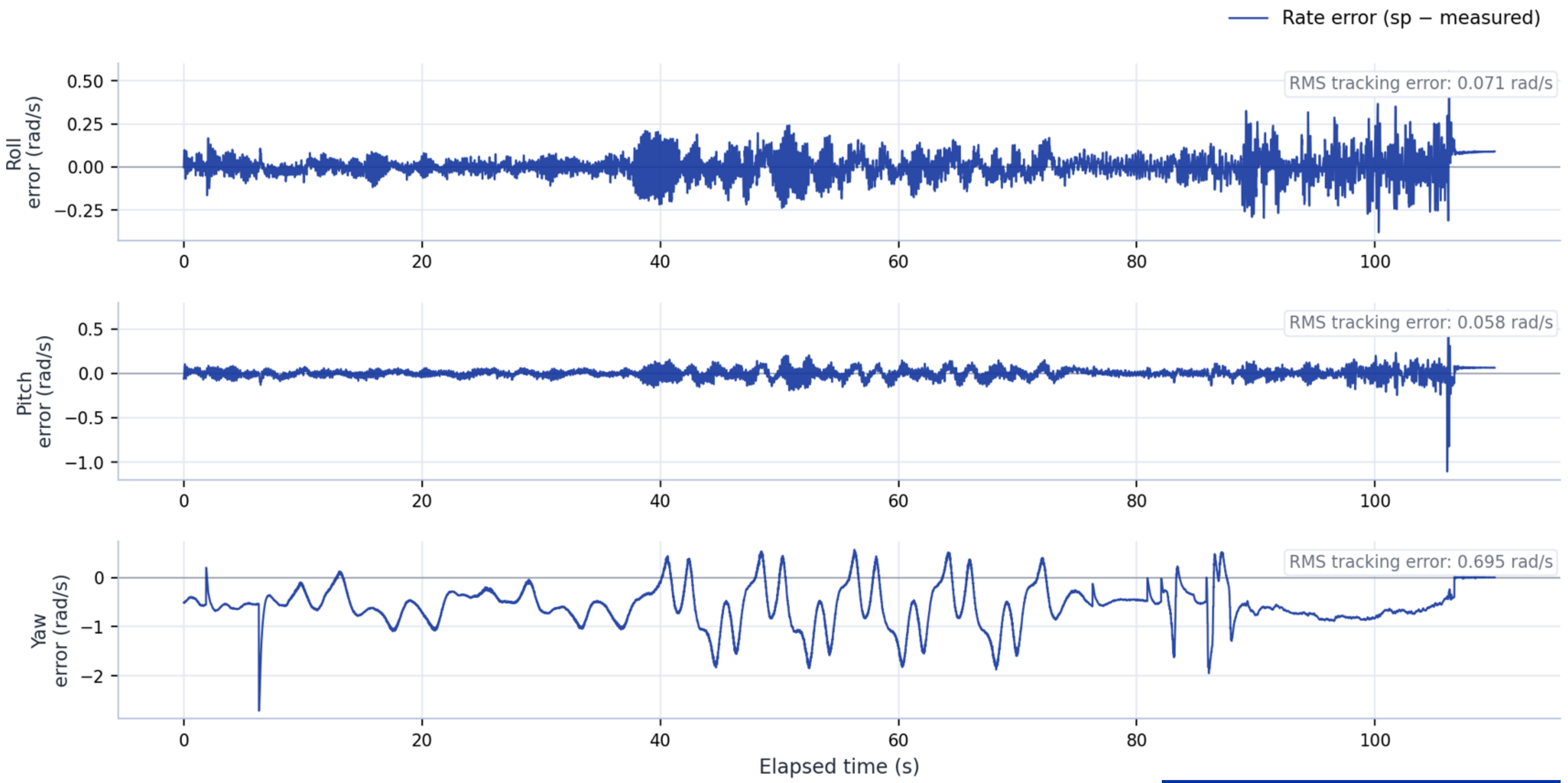
TUNING OBLTR / AAOBLTR

~Zero error between estimated and real states



Aurora FLIGHT SCIENCES
A Boeing Company

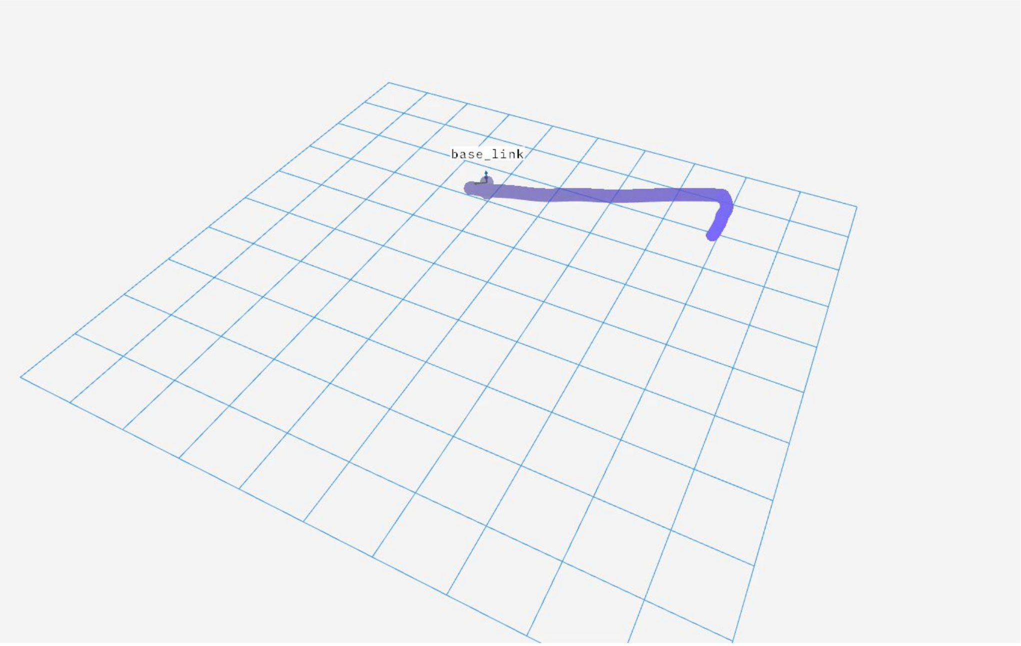
OBLTR / AAOBLTR TRACKING PERFORMANCE



Rates are tracking the setpoint well

Aurora FLIGHT SCIENCES
A Boeing Company

AAOBLTR DOING A FIGURE 8



OBSTACLE AVOIDANCE ANALYTICAL CBF SIMULATION

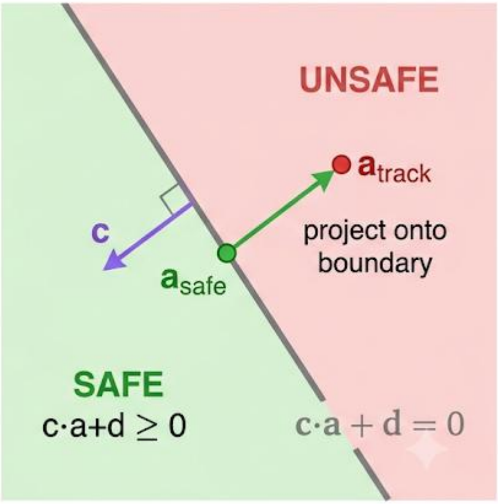
- Closed-form safety filter
- Validated against figure-8 mission with obstacle in path

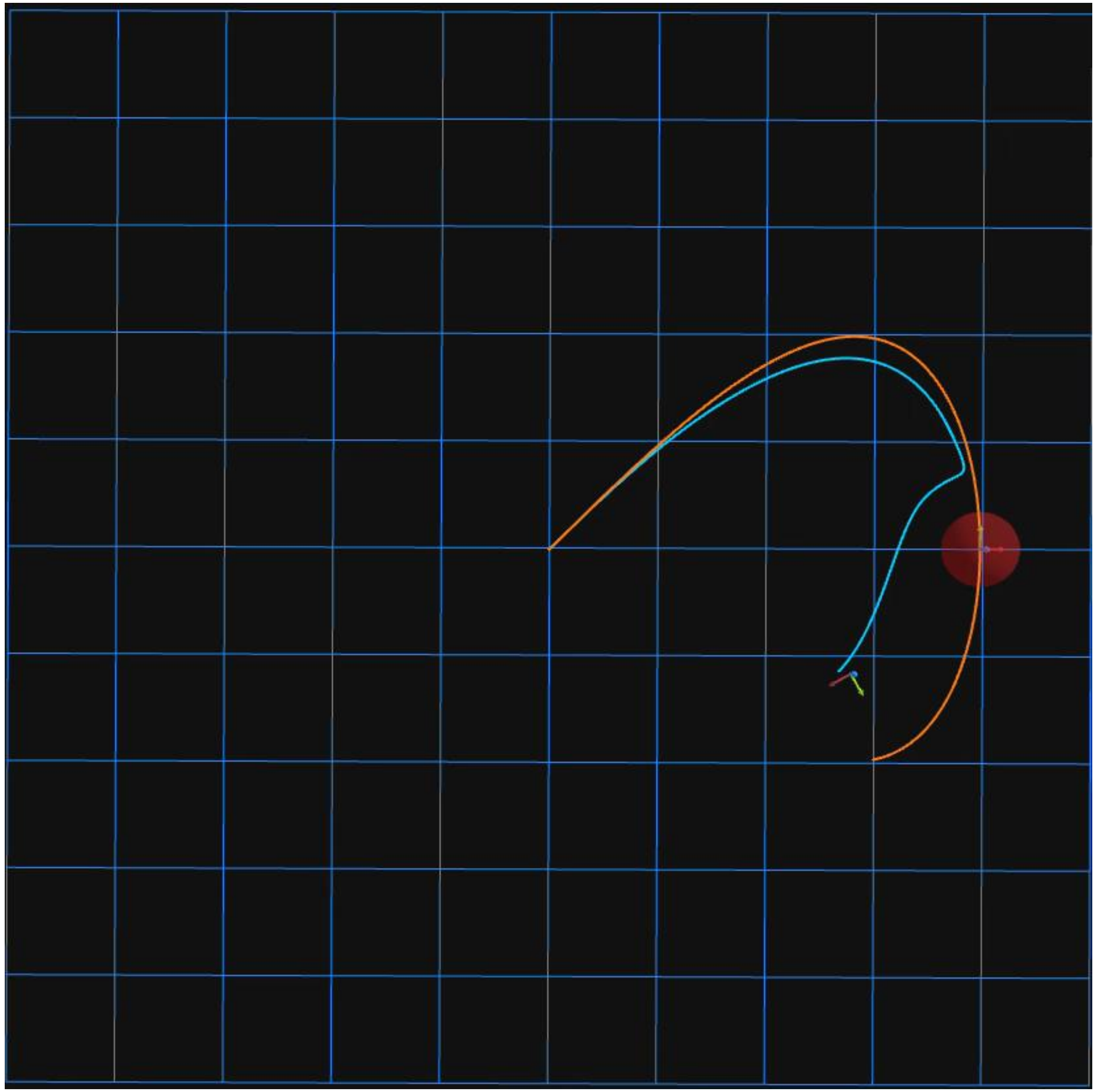
$$\Delta\mathbf{p} = \mathbf{p} - \mathbf{o}$$
$$h = \|\Delta\mathbf{p}\|^2 - r^2$$
$$\psi_1 = 2\,\Delta\mathbf{p}\cdot\mathbf{v} + \alpha_1 h$$
$$c = 2\Delta\mathbf{p}$$
$$d = 2\|\mathbf{v}\|^2 + 2\alpha_1(\Delta\mathbf{p}\cdot\mathbf{v}) + \alpha_2\psi_1$$
$$\text{margin} = c\cdot\mathbf{a}_{\text{track}} + d$$

Solution:

$$\boxed{\begin{aligned} \mathbf{a}_{\text{safe}} &= \mathbf{a}_{\text{track}},\ \text{margin} \geq 0 \\ \mathbf{a}_{\text{safe}} &= \mathbf{a}_{\text{track}} - c(\text{margin}/\|\mathbf{c}\|^2),\ \text{margin} < 0 \end{aligned}}$$

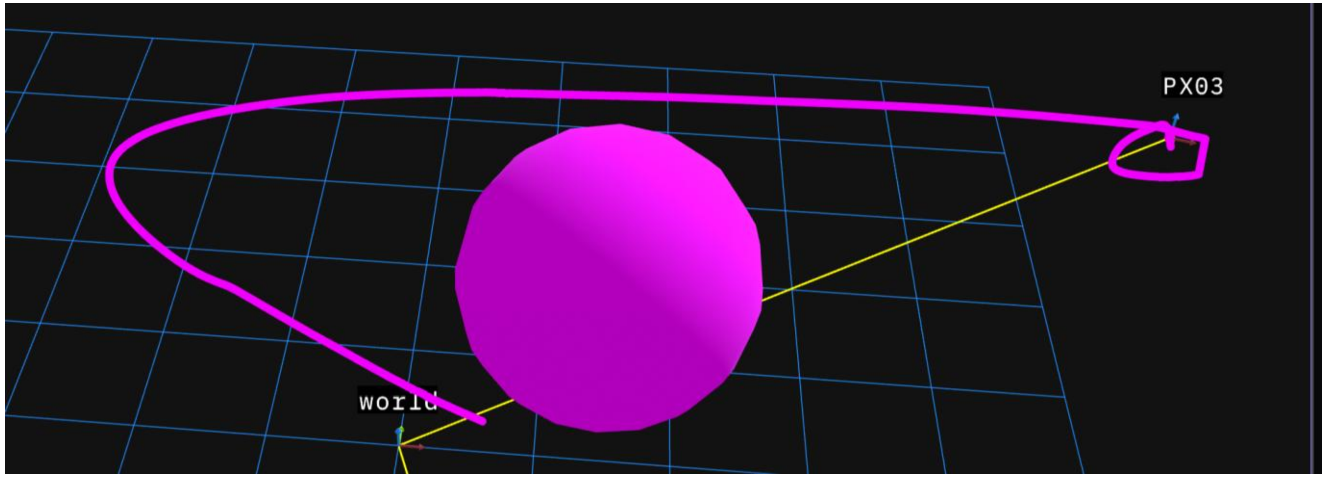
One active obstacle ⇒ one linear inequality in a ⇒ closed form, no QP.





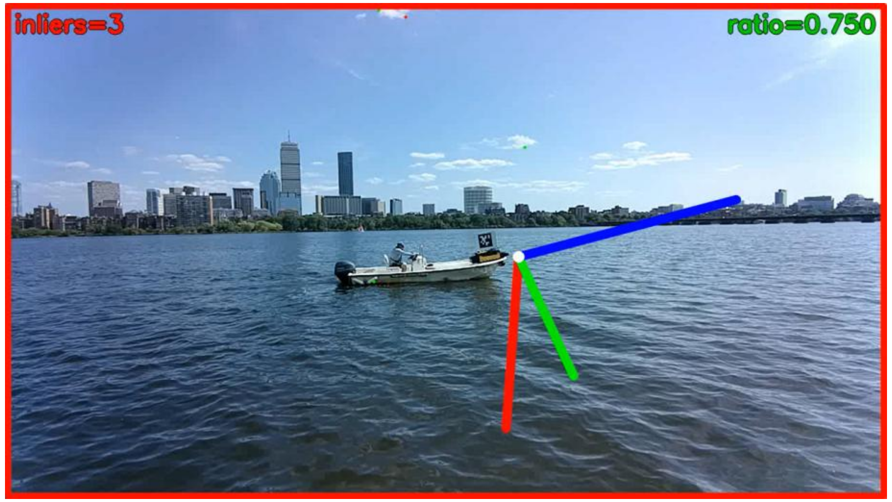
OBSTACLE AVOIDANCE ANALYTICAL CBF HARDWARE

- ROS2 namespace issues blocked goal/CBF message delivery
  - This was due to custom MIT settings
- Algorithm untested on hardware as a result
- Takeaway:
  - Need a more robust HITL pipeline



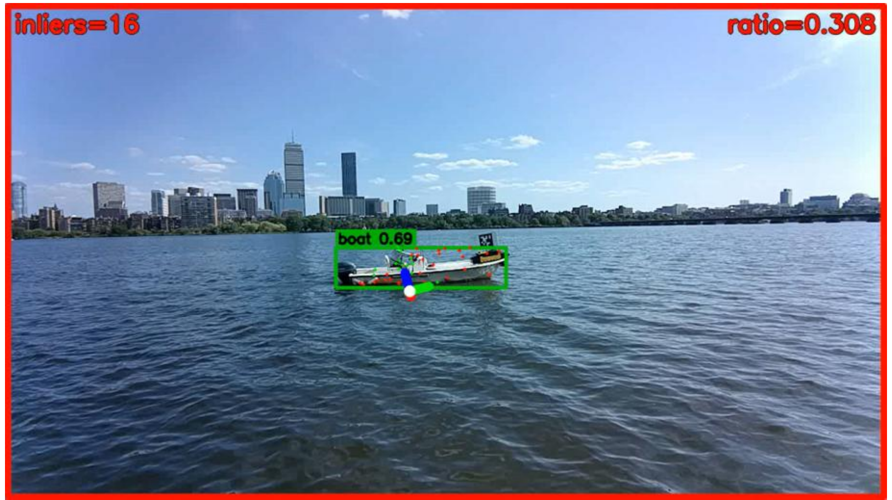
HIERARCHICAL LOCALIZATION STACK

- Vision-based pose estimation for boats without relying on GPS and AprilTags
- Improved HLOC perception stack
  - Apply masking to the observed features
  - Detector: off-the-shelf YOLOv11n
- 2x the detected throughput
- ~10% Improvement in x-axis
- ~2% Improvement in heading
- ~15% Degradation in y-axis



Before

After



LESSONS LEARNED

Technical

- **Hard**ware testing is hard
  - HITL is as import as SITL
  - Building reusable launch tooling to cut setup time
- Steep ramp-up on advanced controllers
- Analyzation tools are essential

Professional

- First time with hard deadlines
- Fast-paced, experienced team

People & Culture

- In-office connections matter
- Great work culture boosts productivity

Thank You To:

Intern Coordinators:

- Katie Martins
- Madelyn Focaracci
- Aidan Green
- Preston Edwards

The FALCON Team:

- Max Greene
- Jeffrey Davis
- Sildomar Monteiro
- Marcel Menner
- Spencer Folk
- Frank Gonzalez
- Trevor Houchens
- Leo Ott

Our Managers:

- Sildomar Monteiro
- Douglas Lipinski

Our Mentors

- Christopher Walter
- Amanda Vollum
- All of AMA

Summary of Contributions

- Extended FALCON controllers and associated missions to quadrotors:
  - RSLQR, OBLTR, AAOBLTR on PX4 firmware
  - Relative station keeping with velocity control
- Designed & implemented analytical CBFs for theoretically guaranteed safe navigation
  - Obstacle avoidance
  - Relative station keeping
- Live DARPA Demonstration: use of FALCON controllers on an aerial platform